

2	(a)	mass is the property of a body resisting changes in motion / quantity of matter in a body / measure of inertia to changes in motion	B1	
		weight is the force due to the gravitational field/force due to gravity or gravitational force	B1	[2]
		Allow 1/2 for ' <i>mass is scalar weight is vector</i> '		
	(b)	(i)	arrow vertically down through O	B1
			tension forces in correct direction on rope	B1 [2]
		(ii)	1. weight = $mg = 4.9 \times 9.81 (= 48.07)$	C1
			$69 \sin \theta = mg$	C1
			$\theta = 44.(1)^\circ$	A1 [3]
			<i>scale drawing allow $\pm 2^\circ$</i>	
			<i>use of cos or tan 1/3 only</i>	
			2. $T = 69 \cos \theta$	C1
			$= 49.6 / 50 \text{ N}$	A1 [2]
			<i>scale drawing $50 \pm 2 (2/2)$ $50 \pm 4 (1/2)$</i>	
			correct answers obtained using scale diagram or triangle of forces will score full marks	
			<i>cos in 1. then sin in 2. (2/2)</i>	